

Report 237

All Fixed Order Asymptotics for Total Kostka Sums

Shifted involutions, margin laws, and controlled inverses

5 October 2026

Abstract

Let a_n be the sum of all entries of the Kostka matrix indexed by partitions of n , the sequence OEIS A104779. We prove, for every fixed $S \geq 0$,

$$a_n = C \sum_{s=0}^S H_s I_{n-s} + O_S(I_n n^{-(S+1)/2}),$$

where I_n is the involution number, $C = \prod_{j \geq 2} (1 - 1/j!)^{-1}$, and the nonnegative constants H_s are effectively computable, with $H_0 = 1$ and $H_1 = H_2 = 0$. A positive global support bound controls the growing family of cycle types, while analytic coefficient stabilization identifies every fixed support. We give half-integer corrections through n^{-3} , rationally certified forty-decimal intervals for five constants, and a Lambert- W /Newton inverse with a precise two-ceiling qualification. The entire nonunit margin profile converges to independent geometric multiplicities, with sharp total-variation distance $(2H_3/C)n^{-3/2} + O(n^{-2})$. The same method gives factorial-shift expansions for A321652 and A068313. An exact Gaussian half-axis split further supplies separately controlled dominant and alternating recessive sectors, the latter smaller by $e^{-2\sqrt{n}}$. The cycle-index identities are due to Schwob; the involution and special-function asymptotic methods are classical. No complete transseries theory, explicit asymptotic onset, or certified finite-input inverse is asserted.

[write] **Status and trust boundary, added 7 October 2026, batch 113.** This is bundle Report 237 of an external AI-assisted research session, filed in ProveIt as the single-source research report `a104779-total-kostka-sums` (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.2). It is unrefereed and not formalized. Its title page and PDF author field read “Report 237”: it names no person or tool. *What rests on what.* The exact identities (2.2) and (9.1) are imported from Schwob’s arXiv version 1 and are not reproved; the write did not read that paper, but its own finite computations confirm both identities for $n \leq 11$ against Kostka numbers computed without the cycle index. Every asymptotic statement has a conventional proof in the text: a positive global support majorant, analytic coefficient stabilization, the classical involution saddle, Gaussian half-axis moments, Stirling’s formula and the mean value theorem. No proof uses a computation; the rational interval certificates of Section 4 use the proved tail bounds. The write adds Remarks 7.3 and 10.1 and dated notes; no statement, proof or number of the source was changed.

Contents

1	The counting model and main results	3
2	The exact cycle index and support constants	7
3	Global support control and stabilization	8
4	Effective coefficients and certified constants	11
5	The involution saddle at every fixed order	13
6	Explicit Kostka sum corrections through cubic order	15
7	Controlled inversion and integer thresholds	16

8	The limiting nonunit margin profile	19
9	Factorial scale companions and binary probability	21
10	Reproducible computations and their limits	23
11	Further questions	25
A	An exact dominant and alternating sector decomposition	26

1 The counting model and main results

For partitions $\lambda, \mu \vdash n$, let $K_{\lambda\mu}$ be the ordinary Kostka number: the number of semistandard Young tableaux of shape λ and content μ . Define

$$a_n = \sum_{\lambda \vdash n} \sum_{\mu \vdash n} K_{\lambda\mu}, \quad a_0 = 1. \quad (1.1)$$

This is OEIS A104779 [1]. The former entry A178718 is officially a duplicate, with offset 1 instead of 0; its positive indices agree with those of A104779, without an index shift [2]. The initial values, at $n = 0, 1, \dots$, are

$$1, 1, 3, 7, 21, 57, 182, 565, 1931, 6670, 24537, 92337, 364602, \dots$$

An equivalent model counts symmetric nonnegative integer matrices, of variable dimension, whose total entry sum is n , whose rows are nonzero, and whose row sums are weakly decreasing. Positions remain distinct when margins coincide. In particular, simultaneous permutations among equal-margin rows and columns are *not* identified. Symmetry makes the column conditions automatic. The empty matrix gives $a_0 = 1$.

All logarithms are natural. Write I_n for the number of involutions in the symmetric group S_n , with

$$I_0 = I_1 = 1, \quad I_n = I_{n-1} + (n-1)I_{n-2} \quad (n \geq 2), \quad (1.2)$$

and put

$$C = \prod_{j \geq 2} (1 - 1/j!)^{-1} = 2.5294774720791526481801161542539542411787 \dots \quad (1.3)$$

The product converges because $\sum_{j \geq 2} 1/j! < \infty$.

Theorem 1.1. *There are nonnegative, effectively computable constants H_s , $s \geq 0$, with $H_0 = 1$ and $H_1 = H_2 = 0$, such that for every fixed integer $S \geq 0$,*

$$a_n = C \sum_{s=0}^S H_s I_{n-s} + O_S(I_n n^{-(S+1)/2}) \quad (n \rightarrow \infty). \quad (1.4)$$

The shifted terms are used only for $n \geq S$. Each constant is specified by the convergent coefficient product in Section 2, and has a finite rational approximation algorithm with a proved error bound. In particular,

$$\frac{a_n}{C I_n} = 1 + H_3 n^{-3/2} + (H_4 - \tfrac{3}{2} H_3) n^{-2} + O(n^{-5/2}), \quad (1.5)$$

where $H_3 = 0.9887340176 \dots$ and $H_4 = 6.0121093539 \dots$

Combining this result with the classical involution expansion gives a Poincaré expansion at every fixed half-integer order. Its leading equivalent is

$$a_n \sim \frac{C}{\sqrt{2}} \left(\frac{n}{e} \right)^{n/2} \exp(\sqrt{n} - 1/4). \quad (1.6)$$

Section 6 supplies all terms through n^{-3} in the multiplicative and logarithmic normalizations. The absence of relative orders $n^{-1/2}$ and n^{-1} in (1.5) is specific to normalization by the *exact* involution number. Those orders are present if the leading equivalent in (1.6) is used instead.

There are two further consequences. Under the uniform distribution on the matrices counted by a_n , the multiplicities of margins $j \geq 2$ converge jointly, in total variation on the entire profile space, to independent geometric variables with parameter $1/j!$. The sharp leading error is $(2H_3/C)n^{-3/2}$. For the nonnegative and binary rectangular sorted-margin models, A321652 and A068313, the analogous expansion has scale $C^2n!$, integer inverse powers, and opposite first relative corrections at order n^{-2} .

1.1 Scope and attribution

Schwob’s exact enumeration of self-inverse double cosets supplies both the Kostka-sum root-count identity and the cycle-index product used here [5, §4.3, (4.5)–(4.8)]. The companion identities are also in that source. These are prior results. The involution saddle method and all-order expansion have classical antecedents in Moser–Wyman [6, 7]; Bornemann [8, Appendix C] provides a modern display of the initial coefficients. We give the analytic details needed for a self-contained application, without claiming the underlying method as new.

The contribution developed here is the global support estimate and stabilization argument for this aggregate, its computable coefficient family, and the consequences for inversion, full margin profiles, and factorial-scale companions. Source comparison is bounded to the actual public versions inspected. In particular, Schwob’s arXiv version 1 was inspected, while the journal full text was inaccessible; no claim about an unexamined journal revision is made. Neither a missing formula in an OEIS entry nor a bounded literature search proves global priority.

The phrase “all fixed orders” means that the truncation order is fixed before $n \rightarrow \infty$. Appendix A strengthens the algebraic result with an exact half-axis decomposition and separate fixed-order expansions for its dominant and alternating recessive sectors. It transfers a classical Gaussian-moment and parabolic-cylinder structure through the proved support estimates. We do not prove convergence of the infinite asymptotic series, uniformity for growing truncation order, optimal truncation, or a complete description of additional exponential effects, Stokes data, or resurgence. Coefficient certificates are effective; the constants and onsets in the asymptotic remainders are not numerically certified. This distinction remains essential when rounding an inverse.

1.2 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 113.** *Provenance.* This report is Report 237 of the session bundle that ProveIt’s docs/incoming intake processed as batch 113: the archive **Report237.zip** (633,646 bytes, 33 files, wrapper directory **Report237/**), arrival commit **60f54ea06**, placed in **a4186a946**. The text is the delivered **article.tex** (55 lines) with its thirteen **\input** files under **sections/** (25 pages, dated 5 October 2026), shipped under the same names. The programs are shipped under **code/** (the delivered root **build.py** as **code/build.py**); the four receipts, delivered in **code/**, and the requirements under **data/**; **COMPUTATION.md** and **SOURCES.md** as delivered. Not shipped, and retrievable from the arrival commit: the delivered **Report237.pdf**, the checksum manifest **MANIFEST.sha256** (32 entries, all verified at the write) and the delivered **README.md**, which the report README replaces. The package records no ProveIt commit; it carries no “prepared for private review” line (**SOURCES.md** only states that it contains no private review notes) and no personal data. It is a single-source report, so no merge choice was made; it is not a Part of **a138178-symmetric-packed-matrices** (relation below). The write prefixed the 123 delivered labels with **tk:** and updated the references to them (64 **\eqref**, 18 **\ref**); it added this subsection, the notes marked [write] and Remarks 7.3 and 10.1, each the last statement of its section, and inserted no numbered display, so every section, theorem and equation of the source keeps its delivered number. One preamble line sets the bibliography ragged right (**etoolbox**), which removes the one underfull line of the delivered build (the entry

of Schwob).

The sources, as the write read them (7 October 2026): OEIS A104779, A178718, A321652 and A068313 in the internal format, with their b-files (Remark 10.1); the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (p0:def:model, p0:thm:lambert-core, p0:def:three-inverses, p0:thm:staircase, plt:def:lw-monomial-log-datum, plt:thm:lw-template). Not read by the write: Schwob (arXiv and journal), Moser–Wyman, Bornemann, Olver and the DLMF; the source’s account of them (Section 1 and SOURCES.md) is reported, not confirmed, in particular the locators §3.1, §4.3 and (4.5)–(4.11) in Schwob’s paper.

What was checked at intake. The batch-113 dossier read the manuscript, found no demonstrably wrong claim, recorded Schwob’s identities as imported, decided that the report is not a Part of `a138178-symmetric-packed-matrices`, and reran the delivered suite on a copy (it reproduces). *What the write checked.* It read every proof and found no error. With its own code (Python 3.14.4, exact integers and fractions; SymPy 1.14.0; mpmath 1.3.0 at 30–60 digits), not importing the delivered programs, it

- computed all Kostka numbers for $n \leq 14$ by horizontal-strip branching and from them a_n , and by the Robinson–Schensted–Knuth and dual correspondences $b_n = \sum_{\lambda} (\sum_{\mu} K_{\lambda\mu})^2$ and $c_n = \sum_{\lambda} (\sum_{\mu} K_{\lambda\mu})(\sum_{\nu} K_{\lambda'\nu})$: equal to the b-files of A104779, A321652 and A068313; counted symmetric matrices with sorted margins directly for $n \leq 7$ and checked (10.1) for $n \leq 8$; evaluated Schwob’s identities (2.2) and (9.1) with its own cycle-index code for $n \leq 11$ (all three agree); checked the square-root formula (2.4) against a brute-force count for every cycle type of size at most 7;
- recomputed C and $H_3, \dots, H_6$ twice, by its own extraction from the product (2.6) at $J = 60$ and by (4.5)–(4.8) (agreement to 50 digits): all five lie in the printed intervals (note after the table in Section 4); $H_2 = 0$, $q_{(2)} = \eta_2$, $q_{(4)} = D_4/4$, $q_{(2,2)} = H_4/2$, $C_0 < 6$, $\mathbb{E}_{\pi} T_3 = H_3$ and $\mathbb{P}_{\pi}(T_3 = 0) = 2/C$; the bounds of Lemma 3.1 for $k < 3000$;
- implemented the recursion (5.3): the coefficients (5.5) are as printed, and with exact I_n for $n = 1000, 4000, 20000$ the remainders of (5.2) and (5.7) after the printed terms, times r^7 and $n^{7/2}$, are bounded (−0.26 and 0.033); derived (5.9), (5.11), (6.1), (6.2) and (6.5) symbolically from the α_j (all equal);
- checked the inverse formulas (7.8)–(7.10) and (9.16) on the model at $n = 10^3, \dots, 10^9$ (each remainder, times its stated scale, stays bounded) and the Newton error orders of Proposition 7.1;
- compared a_n for $15 \leq n \leq 39$ (b-file) with the two-sector expansion (A.5) (note at the end of Appendix A); checked (A.2) and (A.7) by quadrature, and (A.6); compared b_n, c_n for $n \leq 37$ with (9.8) and (9.9) (consistent, not yet in the asymptotic regime).

It reran, on a fresh copy of the delivered layout, `exact_counts.py`, `certify_constants.py`, `symbolic_coefficients.py`, `diagnostics.py` and `sector_diagnostics.py` (normal and -0): the three receipts equal the shipped ones as JSON (on Windows the standard output has CRLF line ends). The guard suite stops on Windows with “backslash output components are forbidden”, a path test, not a mathematical failure. The PDF and ZIP builder and `reproduce_zip.py` were not run. Floating computations are observations, not certificates.

Relation to the repository. Before batch 113 the repository named A321652 and A178718 only in `a260700-parabolic-double-cosets`, which cites Schwob’s paper as studying these aggregates and draws no asymptotic for them; this report gives their asymptotics (A178718 being A104779). The report `a138178-symmetric-packed-matrices` treats symmetric matrices on the same involution scale and names decreasing-margin variants among its further questions; the present aggregate needs a different transform (Schwob’s cycle index, no geometric transform), has its own constants, and adds two rectangular sequences on the factorial scale, so it is a separate report. Neither result is shared; reciprocal notes for both reports are proposed, not applied. No Lean or Rocq file treats these sequences; no statement of this report is formalized, and its place in the collection confers no formal status. The inverse is compared with the transseries volume in Remark 7.3.

[write] **What is not claimed, collected from the source.** Schwob’s cycle-index and Young-subgroup

identities and the companion identities are prior; the involution saddle and its all-order expansion are classical (Moser–Wyman; Bornemann for the displayed coefficients); the parabolic-cylinder sectors are classical (DLMF, Olver). No global novelty from a bounded search or a missing OEIS formula; the journal version of Schwob was not compared. Not claimed: convergence of the asymptotic series, uniformity in the order, optimal truncation, a complete description of exponentially small effects, Stokes data or resurgence; numerical constants or onsets in the remainders; a certified finite-input inverse or rounding decision; that a fixed dominant truncation isolates the recessive sector; moment convergence of the dimension deficit; unrestricted arbitrary-order software; the official offset of A068313 at $n = 0$. *The write adds*: its checks are exact or floating computations; its transseries remark claims no novelty for any inversion.

Reading conventions (letters the source uses in more than one sense, with the tempting false readings; no symbol was renamed).

Symbol	Meanings
$A, a_n, \mathcal{A}_n^\sigma$	a_n the Kostka totals; A the threshold in Section 7; $\mathcal{A}_n^\sigma$ the sectors. <i>False reading</i> : A is not a constant.
B	$B(n)$ the elementary leading function (5.6), $B_\sigma(n)$ its sector versions; B the envelope constant in (7.11); $B_{n,s}$ the support coefficients (3.3); B_{2k} Bernoulli numbers. <i>False reading</i> : the B of (7.11) is not $B(n)$.
C	$C = \prod_{j \geq 2} (1 - 1/j!)^{-1}$; C_0, C_J its majorant and truncations.
F, G	$F(p)$ the cycle-index product, $F(x, t)$ its marked version, F_K the inverse model (7.2); $G(x, t), G(x, u)$ products and $G(x) = \frac{x}{2}(\log x - 1)$ in Section 7.
H, h	H_s the support constants; $h_j(p)$ and $\mathcal{H}_j(x, t)$ cycle-index polynomials.
J, K	$J_s^\pm$ the companion constants; J a product cutoff and the order in Proposition 5.1; $K_{\lambda\mu}$ Kostka numbers, K an order, K_n the matrix dimension, $\mathcal{K}(t)$ the full series.
L, ℓ	$L = \log A$, L_c ; ℓ a cycle length, $\ell(\nu)$ the number of parts, and $\ell = \log x$ in (7.8).
$M, P, \mathbb{P}$	M_j margin multiplicities, $M(\mu)$ fixed-margin counts (10.1), $\mathcal{M}_m^\sigma$ half-axis moments; $P(u)$ the product (2.6), P_n the profile law; $\mathbb{P}$ probability.
q, r, R	q_ρ product coefficients, $q_n = I_{n-3}/I_n$, q a Newton step; $r(\nu)$ square-root counts and r the involution saddle (5.1); R a radius, R_s^* , $R(t)$, R_k , R_k^σ ratios. <i>False reading</i> : q_n is not a q_ρ .
ρ, s, σ	ρ a cycle type without fixed points, and ρ_K the model root of Section 7; s the support size and the shift in (5.11); σ a permutation and the sector sign.
t, T	$t = n^{-1/2}$; $t, t_0 = 1/4$ the marking variable of (3.1); T a support cutoff; T_3 the margin statistic.
α, γ, δ	α_ν cycle-index coefficients; $\alpha_j, \alpha_j^\sigma$ involution coefficients; γ_j saddle coefficients; δ_j companion log coefficients, δ_j in (4.1)’s proof. <i>False reading</i> : α_j is not α_ν .
D, W, N	D_k the sums (4.4) and D the limiting deficit; $W(m)$ the profile weight and W_0 Lambert’s function; $N(A)$ the threshold and $N = \sqrt{n}$ in the appendix.

[write] **Independent check of the write, 7 October 2026.** An independent adversarial check of the batch-113 write (c95388b8e) and of its reciprocal notes (0603ceb9c) used its own code. It read A104779 (#14), A178718 (#60), A321652 (#18) and A068313 (#25) again. The quotations of Remark 10.1 are verbatim, the four b-files have the stated ranges and authors, and A178718 equals A104779 at every positive index.

- *From the OEIS definitions, without Kostka numbers or RSK.* The check counted symmetric matrices with partition margins for $n \leq 16$, and nonnegative and binary matrices with partition margins for $n \leq 11$. They equal the b-files of A104779, A321652 and A068313. It also confirmed the root formula (2.4) by brute force for sizes up to 7, and Schwob’s identity (2.2) in its Young-subgroup form for $n \leq 10$.

- *Constants.* C was recomputed at 50 digits, and $H_3, \dots, H_6$ by its own expansion of (2.6) (H_3 also as $\frac{1}{3} \sum_{j \geq 3} (j)_3 / (j! - 1)$). All five lie in the printed intervals, and the 42 printed decimals of the note in Section 4 are truncations.
- *Numbers.* The inverse formulas (7.8), (7.9) on the model F_4 , and the companion ratios for $n \leq 37$, agree with the write's values. So do the two-sector remainders for $15 \leq n \leq 39$. For these the check adds a precision, a third exponentially small effect: the bracketed note at the end of Appendix A.
- *Reciprocal notes.* Brute force over S_n , $n \leq 6$, confirms the double-coset readings of A104779 (self-inverse $S_\mu w S_\mu$) and A321652 (all $S_\mu w S_\nu$), as stated in `a260700-parabolic-double-cosets`. For `a138178-symmetric-packed-matrices`, the decreasing-margin matrices are among the packed ones, $a_n \leq A_n$ for $n \leq 10$ with A_n recounted, and $a_n/A_n \sim 2Ce^{-L^2/4}L^{n+1}$ is the quotient of the two leading terms.

Also confirmed: Remark 7.3 against the volume's statements (both seeds are `p0:thm:lambert-core` with $w = \log x_0 - 1$); the provenance (33 files, 55 lines, 25 pages, 32 manifest entries); the byte identity of the staged files; the README listing; and the numbering, with 123 delivered labels and 0 differences against the `.aux` of a build of the placed text. No claim of the write was found wrong; one precision was added.

2 The exact cycle index and support constants

For a partition ν with $m_\ell(\nu)$ parts equal to ℓ , define

$$z_\nu = \prod_{\ell \geq 1} \ell^{m_\ell(\nu)} m_\ell(\nu)!, \quad p_\nu = \prod_{\ell \geq 1} p_\ell^{m_\ell(\nu)}, \quad h_j(p) = \sum_{\nu \vdash j} \frac{p_\nu}{z_\nu}.$$

Let $r(\nu)$ be the number of square roots in $S_{|\nu|}$ of a fixed permutation of cycle type ν . This is a square-root count, not a count of involutions commuting with that permutation. With

$$F(p) = \prod_{j \geq 1} (1 - h_j(p))^{-1}, \quad \alpha_\nu = [p_\nu] F(p), \quad (2.1)$$

Schwob's identity is

$$a_n = \sum_{\nu \vdash n} \alpha_\nu r(\nu) = \sum_{\mu \vdash n} \frac{1}{|S_\mu|} \sum_{\sigma \in S_\mu} r(\sigma). \quad (2.2)$$

Here S_μ is the standard Young subgroup, a product of symmetric groups on the blocks specified by μ . All coefficients of F are nonnegative. Formula (2.2) is imported from [5, §4.3]; it fixes the normalization of the entire argument.

2.1 Square roots and nonfixed support

Write $\nu = 1^{n-s}\rho$, where ρ has no parts equal to 1 and $|\rho| = s$. Then

$$r(1^{n-s}\rho) = I_{n-s} r(\rho). \quad (2.3)$$

Indeed, a square root commutes with its square and therefore preserves the fixed-point set of its square. On that set it is an arbitrary involution; on the complement it is a square root of the permutation of type ρ .

For later estimates and computations, the contribution of m cycles of length ℓ to $r(\nu)$ is

$$\begin{cases} \sum_{k=0}^{\lfloor m/2 \rfloor} \frac{m! \ell^k}{(m-2k)! 2^k k!}, & \ell \text{ odd}, \\ 0, & \ell \text{ even and } m \text{ odd}, \\ \frac{m! \ell^{m/2}}{2^{m/2} (m/2)!}, & \ell \text{ even and } m \text{ even}. \end{cases} \quad (2.4)$$

The factors are multiplied over ℓ . One may pair two cycles of the same length, with ℓ possible interlacings; an unpaired odd cycle has a unique root of the same length, while an even cycle cannot be left unpaired. This also explains the source formula in [5, §3.1].

Let u_ℓ have weight ℓ for $\ell \geq 2$, and write $u_\rho = \prod_{\ell \geq 2} u_\ell^{m_\ell(\rho)}$. For $j \geq 2$ set

$$d_j(u) = \sum_{\substack{\rho \text{ has parts } \geq 2 \\ 0 < |\rho| \leq j}} \frac{(j)_{|\rho|}}{(j! - 1) z_\rho} u_\rho, \quad (j)_s = \frac{j!}{(j-s)!}. \quad (2.5)$$

Define the convergent fixed-weight coefficient product

$$P(u) = \prod_{j \geq 2} (1 - d_j(u))^{-1}, \quad q_\rho = [u_\rho] P(u), \quad (2.6)$$

and put

$$H_s = \sum_{\substack{\rho \vdash s \\ \text{parts} \geq 2}} r(\rho) q_\rho, \quad H_0 = 1. \quad (2.7)$$

The empty coefficient of P is 1. The sum for H_1 is empty, and the only type at support 2 is (2), which has no square root. Hence $H_1 = H_2 = 0$.

The notation q_ρ will also be useful for the companions. In particular, $q_{(2)}$ is positive, although $H_2 = 0$. Keeping these two quantities separate avoids confusing a cycle-index coefficient with its root-weighted sum. Absolute convergence, finite computation, and coefficient stabilization are proved next.

3 Global support control and stabilization

The main difficulty is that the nonfixed support s is not bounded as n grows. Fixed-support calculations alone do not justify summing the asymptotic expansion. We first obtain a uniform positive majorant for every support, then prove sharper behavior at each fixed support.

3.1 A uniform cycle index majorant

Substitute $p_1 = x$ and $p_\ell = x^\ell t^\ell$ for $\ell \geq 2$. The homogeneous cycle-index polynomial becomes

$$\mathcal{H}_j(x, t) = h_j(x, x^2 t^2, x^3 t^3, \dots) = x^j [z^j] \frac{\exp((1-t)z)}{1-tz}. \quad (3.1)$$

Although the last expression contains $1-t$, its coefficients in x, t are nonnegative by the original cycle-index definition. Set

$$F(x, t) = (1-x)^{-1} G(x, t), \quad G(x, t) = \prod_{j \geq 2} (1 - \mathcal{H}_j(x, t))^{-1}, \quad B_{n,s} = [x^n t^s] F(x, t).$$

Thus $B_{n,s} = \sum_{\rho \vdash s, \min \rho \geq 2} \alpha_{1^{n-s}\rho}$. At $t_0 = 1/4$,

$$\sum_{j \geq 2} \mathcal{H}_j(1, t_0) = \frac{4}{3}e^{3/4} - 2 < 1, \quad C_0 = \frac{1}{3 - \frac{4}{3}e^{3/4}} < 6. \quad (3.2)$$

For nonnegative v_j with sum less than 1, the finite-product inequality $\prod_j (1 - v_j) \geq 1 - \sum_j v_j$ and monotone convergence imply $G(1, t_0) \leq C_0$. Multiplication by $(1 - x)^{-1}$ takes partial sums of the nonnegative x coefficients of G , so

$$B_{n,s} \leq C_0 4^s \quad (n, s \geq 0). \quad (3.3)$$

This estimate is uniform even when s is comparable with n .

3.2 Root and involution bounds

Lemma 3.1. *For $k \geq 1$, $n \geq 1$, and $0 \leq s \leq n$,*

$$\sqrt{k} \leq \frac{I_k}{I_{k-1}} \leq 1 + \sqrt{k}, \quad \frac{I_{n-s}}{I_n} \leq (n)_s^{-1/2} \leq (e/n)^{s/2}. \quad (3.4)$$

If ρ has no 1-parts and $|\rho| = s \geq 1$, then

$$r(\rho) \leq 2^{s/2} s^{s/4}. \quad (3.5)$$

Proof. Writing $R_k = I_k/I_{k-1}$, the recurrence gives $R_k = 1 + (k-1)/R_{k-1}$. Induction proves the first pair of bounds. For the lower one, with $u = \sqrt{k-1}$, use

$$1 + \frac{u^2}{1+u} = u + \frac{1}{1+u} \geq \sqrt{u^2 + 1};$$

squaring leaves the nonnegative difference $u^2/(1+u)^2$. The upper one follows from $R_{k-1} \geq \sqrt{k-1}$. The initial case is $R_1 = 1$.

Multiplying the lower ratio bounds proves the first bound on I_{n-s}/I_n . The mean of the largest s values of $\log k$, $1 \leq k \leq n$, is at least the mean of all n values. Consequently $(n)_s \geq (n!)^{s/n} \geq (n/e)^s$, proving the second bound.

The ratio upper bound also gives $I_m \leq 2^m m^{m/2}$ for $m \geq 1$. The factor in (2.4) at cycle length $\ell \geq 2$ is at most $\ell^{m_\ell/2} I_{m_\ell}$. Hence

$$r(\rho) \leq 2^{\sum m_\ell} \prod_{\ell \geq 2} (\ell m_\ell)^{m_\ell/2} \leq 2^{s/2} s^{s/4},$$

because $\sum m_\ell \leq s/2$ and $\ell m_\ell \leq s$. □

Combining (3.3), (2.3), and Lemma 3.1, the normalized contribution at support $s \geq 1$ is at most

$$\begin{aligned} \frac{I_{n-s}}{I_n} \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} \alpha_{1^{n-s}\rho} r(\rho) &\leq C_0 4^s 2^{s/2} s^{s/4} (e/n)^{s/2} \\ &\leq C_0 (4\sqrt{2}e n^{-1/4})^s. \end{aligned} \quad (3.6)$$

For all sufficiently large n the base of this geometric bound is less than 1. Thus for every fixed T the total contribution from $s \geq T$ is

$$O_T(I_n n^{-T/4}). \quad (3.7)$$

The estimate is deliberately rough. Its role is uniform control of all large supports, not an attractive numerical onset.

3.3 Exponential stabilization at fixed support

Keep separate marked variables, substituting $p_1 = x$ and $p_\ell = x^\ell u_\ell$ for $\ell \geq 2$. Define

$$G(x, u) = \prod_{j \geq 2} (1 - h_j(x, x^2 u_2, x^3 u_3, \dots))^{-1}, \quad Q_s(x) = \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} r(\rho) [u_\rho] G(x, u).$$

The root-weighted cycle-index coefficient at support s , *before* multiplication by I_{n-s} , is

$$[x^n] \frac{Q_s(x)}{1-x}. \quad (3.8)$$

Lemma 3.2. *For each fixed s and each $1 < R < \sqrt{2}$, the function Q_s is analytic in a neighborhood of $|x| \leq R$, and*

$$[x^n] \frac{Q_s(x)}{1-x} = CH_s + O_{s,R}(R^{-n}). \quad (3.9)$$

Individually, for every fixed type ρ with no 1-parts,

$$\alpha_{1^{n-|\rho|}\rho} = Cq_\rho + O_{\rho,R}(R^{-n}). \quad (3.10)$$

Proof. At $u = 0$ the denominators are $1 - x^j/j!$. For $|x| \leq R < \sqrt{2}$, one has $R^j/j! < 1$ for every $j \geq 2$, with supremum strictly less than 1. Mark all u_ℓ by t^ℓ . For sufficiently small positive t , the positive majorants $\mathcal{H}_j(R, t)$ stay uniformly below 1 and have summable total. To see this, (3.1) gives a finite summed series whenever $Rt < 1$; continuity controls a finite initial set of factors as $t \downarrow 0$, and its summable tail is uniformly small. Only finitely many marked variables can enter a fixed weight. The infinite product is therefore locally uniformly convergent and analytic on a slightly larger x disk and a small polydisc in those variables. Cauchy coefficient extraction proves analyticity of Q_s and of each individual marked coefficient.

The coefficient in (3.8) is the partial sum of the Taylor coefficients of Q_s . The omitted tail is $O_{s,R}(R^{-n})$, so that partial sum tends to $Q_s(1)$. At $x = 1$,

$$h_j(1, u_2, u_3, \dots) = \frac{1}{j!} + \sum_{\substack{\rho \text{ has parts } \geq 2 \\ 0 < |\rho| \leq j}} \frac{u_\rho}{(j - |\rho|)! z_\rho}.$$

Factoring $1 - 1/j!$ from the denominator gives $G(1, u) = CP(u)$, and hence $Q_s(1) = CH_s$. The same argument without the root-weighted sum proves (3.10). $\square$

3.4 Proof of the main theorem

Fix $S \geq 0$ and choose $T = 2S + 2$. At each retained support $s \leq S$, Lemma 3.2 gives $I_{n-s}(CH_s + O(R^{-n}))$. The omitted but bounded supports $S < s < T$ contribute $O_S(I_n n^{-(S+1)/2})$, by the same lemma and (3.4). The entire remaining range $s \geq T$ is bounded by (3.7), with precisely the same order. Exponentially small retained errors can be absorbed. This proves Theorem 1.1.

The second cutoff T is important. Applying the crude global bound directly at $S + 1$ would yield only $n^{-(S+1)/4}$, which is weaker than the natural fixed-support remainder.

4 Effective coefficients and certified constants

At any fixed weight S , the product (2.6) has a finite polynomial approximation. Enumerate partitions with parts at least 2 and size at most S . Truncate each d_j at weight S , replace its reciprocal by

$$(1 - d_j)^{-1} = \sum_{k=0}^{\lfloor S/2 \rfloor} d_j^k \quad \text{modulo terms of weight greater than } S,$$

and multiply for $2 \leq j \leq J$. Every operation uses rational arithmetic. Denote the resulting coefficients by $q_{\rho,J}$ and the root-weighted sums by $H_{s,J}$. Also put

$$C_J = \prod_{j=2}^J (1 - 1/j!)^{-1}, \quad R_s^* = \max_{\substack{\rho \vdash s \\ \min \rho \geq 2}} r(\rho),$$

with $R_0^* = 1$ and an empty maximum interpreted as 0.

Proposition 4.1. *For $J \geq \max(2, s)$,*

$$0 \leq H_s - H_{s,J} \leq 84R_s^* 4^{s-J}, \quad (4.1)$$

$$0 \leq C - C_J \leq \frac{6(J+2)}{(J+1)(J+1)!}. \quad (4.2)$$

For $s \geq 2$ and $J \geq \max(s, 4)$, the stronger fixed-weight estimate is

$$0 \leq H_s - H_{s,J} \leq 24sR_s^* 4^s \frac{(J+1)^s}{(J+1)!}. \quad (4.3)$$

Consequently each H_s is computable to any prescribed absolute precision by a finite algorithm with a proved stopping rule.

Proof. At $t_0 = 1/4$, expansion of (3.1) gives $\mathcal{H}_j(1, t_0) \leq e^3 4^{-j}$. Set

$$\delta_j = d_j(t_0^2, t_0^3, \dots) = \frac{\mathcal{H}_j(1, t_0) - 1/j!}{1 - 1/j!} \leq 2e^3 4^{-j}.$$

Thus $\sum_{j>J} \delta_j < 14 \cdot 4^{-J}$ and $P(t_0) \leq G(1, t_0) \leq C_0 < 6$. Positivity gives

$$P(t_0) - P_J(t_0) = P(t_0) \left(1 - \prod_{j>J} (1 - \delta_j) \right) \leq P(t_0) \sum_{j>J} \delta_j.$$

Bounding a weight- s coefficient by 4^s times this positive total and then multiplying by the maximal root count proves (4.1). The same argument for C , using $C \leq 6$ and

$$\sum_{j>J} \frac{1}{j!} \leq \frac{J+2}{(J+1)(J+1)!},$$

proves (4.2).

For the refinement, truncate each d_j at weight s before evaluating at t_0 . This preserves the desired coefficient and can only decrease the positive product. At each weight k , the sum of $1/z_\rho$

over types without 1-parts is at most 1: it is the proportion of fixed-point-free permutations of k . For $j \geq s$, the truncated factor input is therefore at most $2sj^s/j!$. If $j \geq J+1$ and $J \geq \max(s, 4)$, then

$$\frac{(j+1)^s/(j+1)!}{j^s/j!} = \frac{(1+1/j)^s}{j+1} \leq \frac{e}{J+2} < \frac{1}{2}.$$

The tail is at most $4s(J+1)^s/(J+1)!$. Multiplication by the product bound 6, the coefficient factor 4^s , and R_s^* proves (4.3). $\square$

4.1 The first nonzero support coefficients

Define

$$D_k = \sum_{j \geq \max(2, k)} \frac{(j)_k}{j! - 1}, \quad v_{j,2} = \frac{(j)_2}{2(j! - 1)}, \quad v_{j,3} = \frac{(j)_3}{3(j! - 1)}. \quad (4.4)$$

Direct extraction from (2.6) gives

$$H_3 = D_3/3, \quad (4.5)$$

$$H_4 = D_4/4 + (D_2/2)^2 + \sum_{j \geq 2} v_{j,2}^2, \quad (4.6)$$

$$H_5 = D_5/5, \quad (4.7)$$

$$H_6 = 2D_6/9 + 2(D_3/3)^2 + 2 \sum_{j \geq 3} v_{j,3}^2. \quad (4.8)$$

At support 3 only (3) contributes, with root count 1; at support 4 only (2, 2) contributes, with root count 2; at support 5 only (5) contributes; and at support 6 only (3, 3) contributes, with root count 4. For example, the coefficient of u_2^2 receives both a term from d_j and a term from d_j^2 within a single factor, as well as terms from different factors. These repeated-factor contributions are exactly the extra squared sums in (4.6) and (4.8).

4.2 Forty decimal interval certificates

The following are enclosing intervals, not merely rounded floating-point approximations. Every displayed interval has width 10^{-40} . The finite sums and products use $J = 70$, exact rational arithmetic, and the proved tails (4.2) and (4.3); outward decimal rounding is by integer division. The package reproduces the calculation.

Constant	Certified lower and upper endpoints
C	[2.5294774720791526481801161542539542411787, 2.5294774720791526481801161542539542411788]
H_3	[0.9887340176119067106765900015027659522294, 0.9887340176119067106765900015027659522295]
H_4	[6.0121093539050841781399993190129528805222, 6.0121093539050841781399993190129528805223]
H_5	[0.5456358974558364595261891034594968055729, 0.5456358974558364595261891034594968055730]
H_6	[3.1846595830270382795159539520700360539731, 3.1846595830270382795159539520700360539732]

The constants need not be rational; the finite approximants and the rigorous error bounds are rational. The certification of these constants is logically separate from an effective remainder estimate for a_n : no numerical starting index for an asymptotic theorem follows from this table alone.

[write] **The constants recomputed, 7 October 2026.** The write extracted q_ρ for all types of weight at most 6 from its own truncation of (2.6) at $J = 60$ (exact fractions) and evaluated the closed forms (4.5)–(4.8) at 50 digits; the two agree to 50 digits, and every displayed interval contains the recomputed value:

C	2.529477472079152648180116154253954241178702...
H_3	0.988734017611906710676590001502765952229404...
H_4	6.012109353905084178139999319012952880522219...
H_5	0.545635897455836459526189103459496805572989...
H_6	3.184659583027038279515953952070036053973163...

5 The involution saddle at every fixed order

This section records a classical saddle calculation in a form directly usable with Theorem 1.1. The method and all-order involution asymptotics are prior work [6, 7]. The details below supply a controlled remainder and an explicit finite coefficient rule.

Since $I_n = n![z^n] \exp(z + z^2/2)$, the positive saddle is

$$r = \frac{\sqrt{4n+1} - 1}{2}, \quad n = r^2 + r. \quad (5.1)$$

Proposition 5.1. *For every fixed integer $J \geq 0$,*

$$I_n = \frac{n!e^{r+r^2/2}}{2\sqrt{\pi}r^{n+1}} \left(\sum_{j=0}^J \gamma_j r^{-j} + O_J(r^{-J-1}) \right). \quad (5.2)$$

The rational coefficients are given by the finite recursion

$$\begin{aligned} p_m(v) &= \frac{(iv)^{m+1}}{(m+1)!} + \frac{2^{m+1}(iv)^{m+2}}{(m+2)!}, \quad m \geq 1, \\ Q_0(v) &= 1, \quad Q_j(v) = \frac{1}{j} \sum_{m=1}^j m p_m(v) Q_{j-m}(v), \\ \gamma_j &= \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-v^2} Q_j(v) dv. \end{aligned} \quad (5.3)$$

The integral is an algebraic rule: send odd powers of v to zero and v^{2k} to $(2k-1)!!/2^k$.

Proof. Cauchy's formula on $|z| = r$ gives the prefactor $n!e^{r+r^2/2}/(2\pi r^n)$ times the integral of $e^{\Phi_r(\theta)}$ on $[-\pi, \pi]$, where

$$\Phi_r(\theta) = r(e^{i\theta} - 1 - i\theta) + \frac{1}{2}r^2(e^{2i\theta} - 1 - 2i\theta).$$

Its real part is exactly

$$\Re \Phi_r(\theta) = -r(1 - \cos \theta) - r^2 \sin^2 \theta. \quad (5.4)$$

For $|\theta| \leq \pi/2$ this is at most $-4r^2\theta^2/\pi^2$; for $|\theta| \geq \pi/2$ it is at most $-r$. Fix $0 < \varepsilon < 1/3$. The integral outside $|\theta| \leq r^{-1+\varepsilon}$ is smaller than every fixed algebraic order. The second inequality also controls the possible secondary peak near $\theta = \pi$.

On the central interval put $\theta = v/r$ and $h = 1/r$. Direct Taylor expansion gives

$$\Phi_r(v/r) = -v^2 + \sum_{m \geq 1} h^m p_m(v).$$

The coefficient polynomials of the exponential satisfy (5.3). For fixed J , Taylor's theorem in h bounds the remainder on $|v| \leq r^\varepsilon$ by h^{J+1} times a fixed polynomial in $|v|$ times $e^{\delta v^2}$, with any chosen $\delta > 0$ once r is large. Indeed, derivatives of the analytic exponent of fixed order are bounded by fixed polynomials in $|v|$ when $h|v| = o(1)$, and its non-Gaussian part is bounded by $Ch(v^2 + |v|^3) \leq \delta v^2$. Taking $\delta < 1$ leaves an integrable Gaussian majorant after multiplication by e^{-v^2} . Extending polynomial Gaussian integrals to all of $\mathbb{R}$ adds exponentially small error. Finally $d\theta = dv/r$ gives (5.2). $\square$

The first saddle coefficients are

$$(\gamma_0, \dots, \gamma_6) = \left(1, -\frac{1}{4}, -\frac{7}{96}, \frac{77}{384}, -\frac{3989}{18432}, \frac{3275}{24576}, \frac{1076179}{26542080}\right). \quad (5.5)$$

5.1 Conversion to half integer powers

Write $t = n^{-1/2}$ and

$$B(n) = \frac{1}{\sqrt{2}} \left(\frac{n}{e}\right)^{n/2} e^{\sqrt{n}-1/4}. \quad (5.6)$$

For every fixed J ,

$$I_n = B(n) \left(\sum_{j=0}^J \alpha_j n^{-j/2} + O_J(n^{-(J+1)/2}) \right), \quad (5.7)$$

where

$$\begin{aligned} \alpha_0 &= 1, & \alpha_1 &= \frac{7}{24}, & \alpha_2 &= -\frac{119}{1152}, & \alpha_3 &= -\frac{7933}{414720}, \\ \alpha_4 &= \frac{1967381}{39813120}, & \alpha_5 &= -\frac{57200419}{1337720832}, & \alpha_6 &= \frac{6340449533}{687970713600}. \end{aligned} \quad (5.8)$$

The first three nonconstant coefficients also appear explicitly in Bornemann [8, Appendix C, (120)]. The logarithmic form is

$$\begin{aligned} \log I_n &= \frac{n}{2}(\log n - 1) + \sqrt{n} - \frac{1}{4} - \frac{1}{2} \log 2 \\ &\quad + \frac{7}{24}t - \frac{7}{48}t^2 + \frac{37}{1920}t^3 + \frac{5}{128}t^4 - \frac{5879}{107520}t^5 + \frac{43}{1440}t^6 + O(t^7). \end{aligned} \quad (5.9)$$

For an all-order rule, put $R(t) = \sqrt{1+t^2/4} - t/2$, so that $r = R(t)/t$ and $\log R(t) = -\operatorname{arsinh}(t/2)$. Let $\mathcal{B}(t)$ denote the formal logarithmic correction to $\log B(n)$. Substitution in (5.2) and Stirling's expansion give

$$\begin{aligned} \mathcal{B}(t) &= \frac{R(t)}{2t} + \frac{\operatorname{arsinh}(t/2)}{t^2} + \operatorname{arsinh}(t/2) - \frac{1}{t} + \frac{1}{4} \\ &\quad + \sum_{k \geq 1} \frac{B_{2k}}{2k(2k-1)} t^{4k-2} + \log \left(\sum_{j \geq 0} \gamma_j \left(\frac{t}{R(t)} \right)^j \right). \end{aligned} \quad (5.10)$$

Here B_{2k} are Bernoulli numbers; the apparent negative powers cancel. At any specified order, every sum and series operation is truncated, so the rule is finite. Then $\alpha_j = [t^j] \exp \mathcal{B}(t)$. Proposition 5.1 and Stirling's remainder justify each fixed truncation.

5.2 Ratios at fixed shifts

For any fixed nonnegative integer s , subtraction of sufficiently long logarithmic expansions gives

$$\begin{aligned} \frac{I_{n-s}}{I_n} = n^{-s/2} & \left[1 - \frac{s}{2}t + \frac{3s^2}{8}t^2 + \frac{-7s^3 - 6s^2 + 7s}{48}t^3 \right. \\ & + \frac{25s^4 + 56s^3 - 28s^2 - 56s}{384}t^4 \\ & \left. + \frac{-81s^5 - 340s^4 - 30s^3 + 700s^2 + 111s}{3840}t^5 + O_s(t^6) \right]. \end{aligned} \quad (5.11)$$

For all orders, exponentiate the finite truncation of

$$\begin{aligned} \mathcal{D}_s(t) = \frac{t^{-2} - s}{2} \log(1 - st^2) + \frac{s}{2} + \frac{\sqrt{1 - st^2} - 1}{t} \\ + \mathcal{B}\left(\frac{t}{\sqrt{1 - st^2}}\right) - \mathcal{B}(t). \end{aligned} \quad (5.12)$$

Then I_{n-s}/I_n has asymptotic series $n^{-s/2} \exp \mathcal{D}_s(t)$. Uniformity is asserted only for s in a prescribed finite set. This argument retains enough terms before subtracting; it does not differentiate an unproved remainder.

6 Explicit Kostka sum corrections through cubic order

Combining Theorem 1.1 at $S = 6$ with (5.11) yields, for $t = n^{-1/2}$,

$$\begin{aligned} \frac{a_n}{CI_n} = 1 + H_3 t^3 + (H_4 - \frac{3}{2}H_3)t^4 \\ + (H_5 - 2H_4 + \frac{27}{8}H_3)t^5 \\ + (H_6 - \frac{5}{2}H_5 + 6H_4 - \frac{37}{8}H_3)t^6 + O(t^7). \end{aligned} \quad (6.1)$$

Relative to the elementary leading function $CB(n)$, this becomes

$$\begin{aligned} \frac{a_n}{CB(n)} = 1 + \frac{7}{24}t - \frac{119}{1152}t^2 + (H_3 - \frac{7933}{414720})t^3 \\ + (H_4 - \frac{29}{24}H_3 + \frac{1967381}{39813120})t^4 \\ + (H_5 - \frac{41}{24}H_4 + \frac{3265}{1152}H_3 - \frac{57200419}{1337720832})t^5 \\ + (H_6 - \frac{53}{24}H_5 + \frac{6121}{1152}H_4 - \frac{1453513}{414720}H_3 + \frac{6340449533}{687970713600})t^6 + O(t^7). \end{aligned} \quad (6.2)$$

No coefficients are fitted to sequence data. These expressions are finite compositions of the support coefficients and the rational involution expansion.

For inversion define

$$\kappa_0 = \log C - \frac{1}{4} - \frac{1}{2} \log 2. \quad (6.3)$$

At every fixed order $K \geq 0$ there are effectively computable real coefficients β_j such that

$$\log a_n = \frac{n}{2}(\log n - 1) + \sqrt{n} + \kappa_0 + \sum_{j=1}^K \beta_j n^{-j/2} + O_K(n^{-(K+1)/2}). \quad (6.4)$$

Through $K = 6$ they are

$$\begin{aligned}
\beta_1 &= \frac{7}{24}, & \beta_2 &= -\frac{7}{48}, \\
\beta_3 &= H_3 + \frac{37}{1920}, & \beta_4 &= H_4 - \frac{3}{2}H_3 + \frac{5}{128}, \\
\beta_5 &= H_5 - 2H_4 + \frac{27}{8}H_3 - \frac{5879}{107520}, \\
\beta_6 &= H_6 - \frac{5}{2}H_5 + 6H_4 - \frac{37}{8}H_3 - \frac{1}{2}H_3^2 + \frac{43}{1440}.
\end{aligned} \tag{6.5}$$

The quadratic term $-H_3^2/2$ first appears at order n^{-3} and is required by the logarithm. At any higher fixed order, compute enough H_s , form the ratio series using (5.12), and take a finite logarithmic series. Theorem 1.1 supplies the remainder needed to justify this algebra.

The resulting hierarchy has three useful normalizations. The shifted-involution form preserves the support interpretation and positivity of H_s . The CI_n form displays how the aggregate differs from an exact involution count. The logarithmic form isolates a monotone elementary function suitable for controlled inversion. They are equivalent fixed-order descriptions, rather than independent asymptotic assumptions.

7 Controlled inversion and integer thresholds

There are two different inverse questions. If the input is promised to be an exact sequence value $A = a_n$, one seeks its integer index. For an arbitrary positive threshold A , one seeks

$$N(A) = \min\{n \geq 0 : a_n \geq A\}. \tag{7.1}$$

For large A the equality $a_0 = a_1 = 1$ is immaterial, but a continuous approximation must still be rounded with care.

7.1 The smooth root and Newton iteration

Fix $K \geq 0$ and set

$$\begin{aligned}
G(x) &= \frac{x}{2}(\log x - 1), \\
F_K(x) &= G(x) + \sqrt{x} + \kappa_0 + \sum_{j=1}^K \beta_j x^{-j/2}.
\end{aligned} \tag{7.2}$$

For sufficiently large x ,

$$F'_K(x) = \frac{1}{2} \log x + \frac{1}{2\sqrt{x}} - \sum_{j=1}^K \frac{j\beta_j}{2} x^{-j/2-1} > 0, \quad F''_K(x) = O_K(1/x). \tag{7.3}$$

Let $L = \log A$, and let ρ_K be the unique sufficiently large root of $F_K(\rho_K) = L$.

Proposition 7.1. *If $A = a_n$, then*

$$\rho_K = n + O_K \left(\frac{n^{-(K+1)/2}}{\log n} \right). \tag{7.4}$$

Let W_0 be the principal positive-real Lambert W branch, and set

$$x_0 = \frac{2L}{W_0(2L/e)}. \quad (7.5)$$

Starting from this value, the Newton iterates

$$x_{q+1} = x_q - \frac{F_K(x_q) - L}{F'_K(x_q)} \quad (7.6)$$

satisfy, for each fixed $q \geq 0$,

$$x_q - \rho_K = O_{K,q} \left(x_0^{1-2^{q-1}} (\log x_0)^{-(2^{q+1}-1)} \right). \quad (7.7)$$

Thus finitely many terms and finitely many Newton steps achieve any prescribed fixed algebraic index accuracy asymptotically.

Proof. The expansion (6.4) and the mean value theorem, with $F'_K \asymp \log x$, prove (7.4). No derivative of the unknown remainder is used. The seed solves $G(x_0) = L$ exactly, and $F_K(x) - G(x) = O(\sqrt{x})$. Hence $x_0 - \rho_K = O(\sqrt{x_0}/\log x_0)$, in a neighborhood where $x \asymp x_0$. Taylor's theorem for the explicit smooth function F_K gives the Newton error recurrence

$$e_{q+1} = O_K \left(\frac{e_q^2}{x_0 \log x_0} \right), \quad e_q = x_q - \rho_K.$$

Induction yields (7.7). Its successive orders are

$$O\left(\frac{\sqrt{x_0}}{\log x_0}\right), \quad O((\log x_0)^{-3}), \quad O(x_0^{-1}(\log x_0)^{-7}), \quad O(x_0^{-3}(\log x_0)^{-15}), \dots$$

□

7.2 Explicit initial inverse terms

Center the constant by setting

$$L_c = \log A - \kappa_0, \quad x = \frac{2L_c}{W_0(2L_c/e)}, \quad \ell = \log x.$$

For exact-term inputs $A = a_n$,

$$\begin{aligned} n = x - \frac{2\sqrt{x}}{\ell} + \frac{2}{\ell^2} - \frac{2}{\ell^3} \\ - x^{-1/2} \left(\frac{2\beta_1}{\ell} + \frac{1}{\ell^3} - \frac{14}{3\ell^4} + \frac{4}{\ell^5} \right) + O\left(\frac{x^{-1}}{\ell}\right). \end{aligned} \quad (7.8)$$

Here $2\beta_1/\ell = 7/(12\ell)$. The next term is

$$x^{-1} \left(-\frac{2\beta_2}{\ell} - \frac{4\beta_1}{\ell^3} - \frac{4}{\ell^5} + \frac{40}{3\ell^6} - \frac{10}{\ell^7} \right); \quad (7.9)$$

including it improves the error to $O(x^{-3/2}/\ell)$. Without centering, take $x = 2L/W_0(2L/e)$ and $\ell = \log x$. Then

$$n = x - \frac{2\sqrt{x}}{\ell} - \frac{2\kappa_0}{\ell} + \frac{2}{\ell^2} - \frac{2}{\ell^3} + O(x^{-1/2}/\ell). \quad (7.10)$$

Higher terms follow by substituting $n = x + \sum_{j=-1}^M d_j(\ell)x^{-j/2}$ in $F_K(n) - L$ and cancelling successive powers of $x^{-1/2}$. Every new coefficient enters linearly with nonzero factor $\ell/2$. Consequently the d_j are rational functions of ℓ , with polynomial dependence on κ_0 and the β_j . The package checks the displayed formulas by exact symbolic substitution.

7.3 A precise two ceiling enclosure

The matrix model proves $a_{n+1} > a_n$ for every $n \geq 1$. Append a final row and column of zeros except for a new diagonal 1. Every old margin is at least 1, so weakly decreasing order is preserved, and deletion recovers the original matrix. The one-entry matrix $[n+1]$ is outside the image, which has at least two rows. This proves strict increase; it does not claim strict increase between $n=0$ and $n=1$.

Put $p = (K+1)/2$. By (6.4) there exist constants $B > 0$ and N_0 such that

$$|\log a_n - F_K(n)| \leq Bn^{-p} \quad (n \geq N_0). \quad (7.11)$$

Increase N_0 so that both $F_K(x) + Bx^{-p}$ and $F_K(x) - Bx^{-p}$ are increasing. For sufficiently large $L = \log A$, define x_- and x_+ by

$$F_K(x_-) + Bx_-^{-p} = L, \quad F_K(x_+) - Bx_+^{-p} = L. \quad (7.12)$$

Theorem 7.2. *For every sufficiently large threshold A ,*

$$\lfloor x_- \rfloor \leq N(A) \leq \lceil x_+ \rceil, \quad x_+ - x_- = O_K \left(\frac{x_-^{-p}}{\log x_-} \right). \quad (7.13)$$

Eventually the two displayed ceilings either coincide or are consecutive integers.

Proof. Every integer $n < \lfloor x_- \rfloor$ in the range of (7.11) has $\log a_n < L$ by the upper envelope. Once A is sufficiently large, all smaller integers in the omitted finite prefix also have $a_n < A$. Conversely $n = \lceil x_+ \rceil$ has $\log a_n \geq L$ by the lower envelope. These prove the two inequalities. The mean value theorem, with derivative comparable to $\log x_-$, gives the width estimate. It is eventually less than 1. $\square$

For promised exact terms, any approximation with an eventually proved $o(1)$ absolute index error recovers n by nearest-integer rounding for all sufficiently large n . For arbitrary thresholds, blindly taking $\lceil \rho_K \rceil$ is unjustified near a jump: A may lie arbitrarily close to an exact sequence value. If certified roots and remainder constants are available, the two candidates in (7.13) can be resolved by exact neighboring-term comparison. The present existence-form bounds do not supply numerical values of B, N_0 , a finite-input onset, or a certified rounding decision at any particular input.

Remark 7.3 ([write] The inverse and the transseries volume, 7 October 2026). Compared statement by statement with the repository's transseries volume [11]; the volume's symbols are not used here.

1. *The growth.* $\log a_n = \frac{n}{2}(\log n - 1) + \sqrt{n} + \kappa_0 + O(n^{-1/2})$ by (6.4), with a term $n \log n$ (and the companions likewise by (9.14)), so none of the three sequences is of the exponential–power form of `p0:def:model`; no instance of the volume’s model is claimed.
2. *The Lambert seeds.* With $w = \log x_0 - 1$, the seed equation $G(x_0) = L$ of (7.5) becomes $w + \log w = \log(2L) - 1$: the volume’s dominant block `p0:thm:lambert-core` with unit slope, unit logarithmic coefficient and target $\log(2L) - 1$, whose principal branch gives the displayed x_0 . An exact instance; so is the companion seed (9.15), with target $\log L - 1$.
3. *The model root, the Newton iterates and the explicit inverse terms.* The model F_K adds to the Lambert block the term $\sqrt{x}$ and powers $x^{-j/2}$, which in the chart of $\log x$ are not series in its reciprocal with coefficients polynomial in its logarithm; the expansions (7.8)–(7.10) proceed in powers of $x^{-1/2}$ with coefficients rational in $\log x$. So the write has not shown ρ_K , the iterates or these expansions to be instances of `plt:thm:lw-template`; they are proved directly.
4. *The thresholds.* $N(A)$ is the volume’s staircase (`p0:def:three-inverses(1)`) of a_n , strictly increasing from $n = 1$. The functions $F_K \pm Bx^{-p}$ are smooth and eventually increasing but do not take the values $\log a_n$ at the integers, so they are not admissible interpolations; Theorem 7.2, proved directly at the integers, is an analogue of `p0:thm:staircase(2)`, and the nearest-integer recovery of a promised exact term an analogue of `p0:thm:staircase(3)`, not instances. The companion thresholds of Section 9 are analogues in the same sense.

8 The limiting nonunit margin profile

Choose uniformly a symmetric sorted-positive-margin matrix counted by a_n . For $j \geq 2$, let M_j be the number of its margins equal to j . A profile $m = (m_j)_{j \geq 2}$ has finite support and is feasible when

$$W(m) = \sum_{j \geq 2} j m_j \leq n. \quad (8.1)$$

It then determines the remaining $n - W(m)$ unit margins. This profile refers to distinct row positions in the matrix model, not to a quotient by simultaneous permutations.

Put

$$w(m) = \prod_{j \geq 2} (j!)^{-m_j}, \quad \pi(m) = \frac{w(m)}{C}, \quad T_3(m) = \sum_{j \geq 3} 2 \binom{j}{3} m_j. \quad (8.2)$$

The law π is the product of independent geometric distributions on $\{0, 1, 2, \dots\}$,

$$\mathbb{P}_\pi(M_j = m) = (1 - 1/j!)(1/j!)^m. \quad (8.3)$$

Since $\sum_{j \geq 2} \mathbb{P}_\pi(M_j > 0) = \sum_{j \geq 2} 1/j! < \infty$, it is supported on finite profiles. Its normalization is exactly C , and

$$\mathbb{E}_\pi T_3 = \sum_{j \geq 3} \frac{2 \binom{j}{3}}{j! - 1} = H_3. \quad (8.4)$$

Let P_n be the actual profile law, extended by zero to infeasible profiles, and put $q_n = I_{n-3}/I_n$ for $n \geq 3$.

Theorem 8.1. *On the countable space of finite nonunit margin profiles,*

$$\left\| \frac{dP_n}{d\pi} - 1 - q_n(T_3 - H_3) \right\|_{L^1(\pi)} = O(n^{-2}). \quad (8.5)$$

In particular,

$$d_{\text{TV}}(P_n, \pi) = \frac{2H_3}{C}n^{-3/2} + O(n^{-2}). \quad (8.6)$$

Thus the entire nonunit margin profile converges jointly in total variation to the independent geometric law.

Proof. For a feasible profile, apply the Young-subgroup sum in (2.2) to its single margin partition. The identity permutation contributes exactly $I_n w(m)$. No support-1 or support-2 element contributes. A support-3 element is a single 3-cycle in a block of size j , with $2\binom{j}{3}$ choices per such block. Its square-root count is I_{n-3} . Therefore the number of matrices of profile m is exactly

$$I_n w(m) + I_{n-3} w(m) T_3(m) + R_{n,m}, \quad R_{n,m} \geq 0, \quad (8.7)$$

where the remainder contains supports at least 4. The proof of Theorem 1.1 gives

$$\sum_{W(m) \leq n} R_{n,m} = O(I_n n^{-2}). \quad (8.8)$$

For every $1 < R < \sqrt{2}$, the exponential moment

$$\mathbb{E}_\pi R^W = \prod_{j \geq 2} \frac{1 - 1/j!}{1 - R^j/j!}$$

is finite. The T_3 -weighted moment is also finite, either by differentiating the geometric factors and summing a factorially decaying series, or by bounding T_3 by a polynomial in W and choosing a slightly larger admissible R . Consequently both $\sum_{W > n} \pi(m)$ and $\sum_{W > n} \pi(m) T_3(m)$ are $O(R^{-n})$. Summing (8.7) and using (8.8) now gives

$$\frac{a_n}{CI_n} = 1 + q_n H_3 + O(n^{-2}).$$

Divide (8.7) by $a_n \pi(m)$, sum absolute errors against π , and include the exponentially small infeasible tail. The remainder has L^1 mass $O(n^{-2})$ by positivity. Since $q_n = O(n^{-3/2})$, quadratic normalization terms are $O(n^{-3})$. This proves (8.5).

Total variation is half the L^1 density difference, so

$$d_{\text{TV}}(P_n, \pi) = \frac{q_n}{2} \mathbb{E}_\pi |T_3 - H_3| + O(n^{-2}).$$

The variable T_3 is nonnegative and every positive value is at least 2. The certificate in Section 4 proves $0 < H_3 < 2$. Since its mean is H_3 ,

$$\mathbb{E}_\pi |T_3 - H_3| = 2H_3 \mathbb{P}_\pi(T_3 = 0), \quad \mathbb{P}_\pi(T_3 = 0) = \prod_{j \geq 3} (1 - 1/j!) = \frac{2}{C}.$$

Finally (5.11) gives $q_n = n^{-3/2} + O(n^{-2})$, proving (8.6). $\square$

8.1 The matrix dimension deficit

If K_n is the matrix dimension, then

$$n - K_n = \sum_{j \geq 2} (j-1)M_j.$$

Total variation contracts under a measurable map. Thus the law of this deficit converges, with error at most that in (8.6), to the finite random variable $D = \sum_{j \geq 2} (j-1)M_j$ under (8.3). Its probability generating function is

$$\mathbb{E}z^D = \prod_{j \geq 2} \frac{1 - 1/j!}{1 - z^{j-1}/j!}, \quad |z| \leq 1. \quad (8.9)$$

For example, the limiting probability that every margin equals 1 is $1/C$, and that no margin exceeds 2 is $2/C$. These follow from the joint profile theorem. Distributional and total-variation convergence do not by themselves assert convergence of unbounded moments of the actual finite- n deficit; no such additional assertion is needed here.

9 Factorial scale companions and binary probability

Let b_n count all rectangular nonnegative integer matrices of total entry sum n , with no zero rows or columns and with weakly decreasing row and column sums. Let c_n count the same objects with entries restricted to 0 or 1. These are A321652 and A068313, with official offsets 0 and 1, respectively [3, 4]. We use $b_0 = 1$ and set $c_0 = 1$ as the natural empty-object extension, not as a claim about the official offset of A068313. Dimensions vary, and equal-margin positions remain distinct.

With α_ν as in (2.1), Schwob gives the exact identities

$$b_n = \sum_{\nu \vdash n} \alpha_\nu^2 z_\nu, \quad c_n = \sum_{\nu \vdash n} \alpha_\nu^2 (-1)^{n-\ell(\nu)} z_\nu, \quad (9.1)$$

where $\ell(\nu)$ is the number of parts [5, §4.3, (4.9), (4.11)]. These identities are prior work. They use the same cycle-index coefficients as the Kostka total, with different weights.

For $s \geq 0$, define

$$J_s^+ = \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} z_\rho q_\rho^2, \quad J_s^- = \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} (-1)^{s-\ell(\rho)} z_\rho q_\rho^2, \quad J_0^\pm = 1. \quad (9.2)$$

Theorem 9.1. *For every fixed integer $S \geq 0$,*

$$b_n = C^2 \sum_{s=0}^S J_s^+ (n-s)! + O_S(n!n^{-S-1}), \quad (9.3)$$

$$c_n = C^2 \sum_{s=0}^S J_s^- (n-s)! + O_S(n!n^{-S-1}). \quad (9.4)$$

All coefficients are effectively computable from the same finite product as in Proposition 4.1.

Proof. For $\nu = 1^{n-s}\rho$, one has $z_\nu = (n-s)!z_\rho$ and $(-1)^{n-\ell(\nu)} = (-1)^{s-\ell(\rho)}$. Each fixed-support term therefore has the asserted limit by (3.10). For the growing-support tail, write $\alpha_{n,\rho} = \alpha_{1^{n-s}\rho}$. Then

$$z_\rho = \prod_{\ell \geq 2} \ell^{m_\ell} m_\ell! \leq \prod_{\ell \geq 2} (\ell m_\ell)^{m_\ell} \leq s^{s/2},$$

and positivity with (3.3) gives

$$\sum_{\rho \vdash s, \min \rho \geq 2} \alpha_{n,\rho}^2 \leq \left(\sum_{\rho} \alpha_{n,\rho} \right)^2 \leq C_0^2 16^s.$$

Also $(n-s)!/n! \leq (e/n)^s$. Thus the absolute contribution of support s , normalized by $n!$, is at most

$$C_0^2 (16e n^{-1/2})^s. \quad (9.5)$$

The tail starting at T is $O_T(n^{-T/2})$. Splitting omitted supports at $T = 2S + 2$, as in the main proof, gives $O_S(n^{-S-1})$. The signed c_n tail is bounded in absolute value by the positive b_n tail. $\square$

9.1 Initial coefficients and their interpretation

Set

$$\eta_2 = q_{(2)} = D_2/2 = \sum_{j \geq 2} \frac{\binom{j}{2}}{j! - 1} > 0. \quad (9.6)$$

The separate notation matters: η_2 is not H_2 , which vanishes. The first coefficients are

$$\begin{aligned} J_1^+ &= J_1^- = 0, \\ J_2^+ &= 2\eta_2^2, & J_2^- &= -2\eta_2^2, \\ J_3^+ &= 3H_3^2, & J_3^- &= 3H_3^2, \\ J_4^+ &= D_4^2/4 + 2H_4^2, & J_4^- &= -D_4^2/4 + 2H_4^2. \end{aligned} \quad (9.7)$$

Indeed, $q_{(3)} = H_3$, $q_{(4)} = D_4/4$, and $q_{(2,2)} = H_4/2$; the centralizer factors at the last two types are 4 and 8, with signs minus and plus.

Writing $u_n^+ = b_n$ and $u_n^- = c_n$, conversion of factorial shifts gives

$$\begin{aligned} \frac{u_n^\pm}{C^2 n!} &= 1 + J_2^\pm n^{-2} + (J_2^\pm + J_3^\pm) n^{-3} \\ &\quad + (J_2^\pm + 3J_3^\pm + J_4^\pm) n^{-4} + O(n^{-5}). \end{aligned} \quad (9.8)$$

There is no relative n^{-1} correction. The first deviations from $C^2 n!$ have opposite signs. In particular,

$$\frac{b_n}{c_n} = 1 + 4\eta_2^2 n^{-2} + O(n^{-3}), \quad (9.9)$$

$$b_n - c_n = 4C^2 \eta_2^2 (n-2)! + O(n! n^{-4}). \quad (9.10)$$

The stronger remainder in the difference uses cancellation at support 3. Binary matrices are literally a subset of the same variable-dimension, sorted-margin matrix universe. Therefore a uniformly chosen matrix counted by b_n is binary with probability

$$\frac{c_n}{b_n} = 1 - 4\eta_2^2 n^{-2} + O(n^{-3}). \quad (9.11)$$

The agreement of the row and column conventions is necessary for this probability interpretation.

9.2 Effective companion constants

The coefficient version of (4.3) gives, for $s \geq 2$ and $J \geq \max(s, 4)$,

$$0 \leq q_\rho - q_{\rho,J} \leq E_{s,J}, \quad E_{s,J} = 24s4^s \frac{(J+1)^s}{(J+1)!}, \quad q_\rho \leq 6 \cdot 4^s. \quad (9.12)$$

Since $q_\rho^2 - q_{\rho,J}^2 \leq 12 \cdot 4^s E_{s,J}$, either sign satisfies

$$\left| J_s^\pm - \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} \varepsilon_\rho^\pm z_\rho q_{\rho,J}^2 \right| \leq 12 \cdot 4^s E_{s,J} \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} z_\rho, \quad (9.13)$$

where $\varepsilon_\rho^+ = 1$ and $\varepsilon_\rho^- = (-1)^{s-\ell(\rho)}$. The plus approximation is one-sided; the signed one need not be. This is a finite stopping rule, though not an optimized bound.

9.3 Companion inverses

Stirling's expansion and Theorem 9.1 imply, for every fixed K ,

$$\log u_n^\pm = n(\log n - 1) + \frac{1}{2} \log n + \delta_0 + \sum_{j=1}^K \delta_j^\pm n^{-j} + O_K(n^{-K-1}), \quad (9.14)$$

with

$$\delta_0 = \frac{1}{2} \log(2\pi) + 2 \log C, \quad \delta_1^\pm = \frac{1}{12}, \quad \delta_2^\pm = J_2^\pm, \quad \delta_3^\pm = J_2^\pm + J_3^\pm - \frac{1}{360}.$$

For $L = \log A$, the Lambert- W seed is now

$$x_0 = \frac{L}{W_0(L/e)}, \quad x_0(\log x_0 - 1) = L. \quad (9.15)$$

The smooth truncated logarithm has derivative $\log x + O(1/x)$, so its root has index error $O(x^{-K-1}/\log x)$ for an exact-term input. Its seed error is $O(1)$, and Newton iteration has local error recurrence $e_{q+1} = O(e_q^2/(x_0 \log x_0))$. In particular,

$$n = x_0 - \frac{1}{2} - \frac{\delta_0}{\log x_0} + O\left(\frac{1}{x_0 \log x_0}\right). \quad (9.16)$$

Every fixed precision is obtained by finitely many terms and steps.

Both sequences are eventually strictly increasing because their adjacent ratios are asymptotic to n . For threshold indices, use the upper and lower envelopes obtained by adding and subtracting Bx^{-K-1} from the smooth logarithm, invert them, and take their two ceilings. The resulting root width is $O(x^{-K-1}/\log x)$, so at most two consecutive candidates remain eventually. The same finite-input caveat as in Theorem 7.2 applies: the existence of asymptotic constants does not certify a numerical rounding decision.

10 Reproducible computations and their limits

The accompanying source package contains the complete L^AT_EX article, the PDF, bounded Python programs, exact receipts, a dependency list, public source notes, and a build-and-replay guide. Computations serve three distinct roles.

10.1 Exact arithmetic checks

The finite cycle-index product (2.1), evaluated with rational coefficients, yields the exact values of all three sequences through $n = 20$. The three weightings are respectively $r(\nu)$, z_ν , and $(-1)^{n-\ell(\nu)}z_\nu$. They agree with the stated OEIS prefixes: A104779 and A321652 at $n = 0, \dots, 20$, and A068313 at $n = 1, \dots, 20$. The binary $n = 0$ value is labelled as an extension.

An independent symmetric-matrix recursion checks the Kostka total by prescribed margins. Choose a row of least positive margin, enumerate its off-diagonal entries within the remaining capacities, and place its unused margin on the diagonal. Recurse on the residual symmetric matrix. Equal residual margins are grouped for speed, with multinomial factors restoring every labelled assignment; matrices are never identified under permutation. Summing over partitions of n produces an independent exact count. Two useful normalization identities are

$$M(2, 1^{n-2}) = \frac{I_n}{2}, \quad M(3, 1^{n-3}) = \frac{I_n + 2I_{n-3}}{6}, \quad (10.1)$$

in their natural ranges $n \geq 2$ and $n \geq 3$. They also follow immediately from the Young-subgroup formula for S_2 and S_3 .

Exact finite multivariate coefficients independently recover (4.5)–(4.8). The certificate program computes rational approximants at $J = 70$, adds the proved rational tail bounds, and uses integer floors and ceilings to print the enclosing intervals. Exact symbolic calculations verify the saddle coefficients, logarithmic and multiplicative conversions, shifted ratios, and inverse reversion displayed in the article. These checks do not substitute for the global support estimate, analytic stabilization, saddle remainder, or mean-value arguments.

10.2 Diagnostics and safe reproduction

Floating-point evaluations are kept separate from deterministic exact receipts. A long decimal output or a small numerical residual is a diagnostic, not a directed interval certificate. In particular, comparison against exact involution integers at large finite n cannot certify a uniform remainder constant or an onset for inverse rounding.

The public interfaces are intentionally bounded. Their parameter and path checks raise explicit exceptions, including when Python is run with optimization enabled. No correctness check relies on an assertion. The integer-to-string conversion limit remains enabled, including tests at a cap of 640 digits; large exact integers are not dumped into receipts. Child Python processes use the no-bytecode option. The build uses an exclusive external output directory, leaves the source tree unchanged, and disables shell escape during $\text{\TeX}$ format preparation and compilation. The actual ZIP replay checks member names and metadata, the manifest, regenerated exact receipts, and the reproduced PDF under both ordinary and optimized execution. The package guide states the precise tested commands and ranges.

The bounded programs illustrate and verify the mathematical algorithms at stated orders; they are not advertised as unrestricted arbitrary-order software. The mathematical effectivity assertion in Proposition 4.1 is broader: any *fixed* target weight and precision admit a finite computation with a proved stopping rule. No third-party paper PDF is redistributed.

[write] **The shipped files, 7 October 2026.** In ProveIt the PDF, the manifest and the delivered README are not shipped; the programs are `code/*.py` (the builder as `code/build.py`), and the receipts `code/*_receipt.json` and the requirements are shipped under `data/`. The checkers write to standard output

and run from the shipped files; the builder and the archive verifier expect the delivered layout and the manifest, so they run only in a re-extracted archive. The report README gives commands.

Remark 10.1 ([write] The OEIS entries, 7 October 2026). The write read the entries in the internal format on 7 October 2026. *A104779*, revision #14 (4 September 2025), by Alford Arnold (24 March 2005): “a(n) is the sum of entries of n-th Kostka matrix for the partitions of n.”, with the matrix interpretation by Ludovic Schwob (29 August 2023), a link to Schwob’s paper (“See p. 13”) and a b-file by Schwob for $n = 0, \dots, 39$. *A178718*, revision #60 (4 September 2025), by Wouter Meeussen (26 December 2010): “Duplicate of A104779.”, offset 1, b-file by Schwob for $n = 1, \dots, 39$, equal to A104779 at the same indices, as the source says. *A321652*, revision #18 (11 June 2025), by Gus Wiseman (15 November 2018), and *A068313*, revision #25 (11 June 2025), by Axel Kohnert (25 February 2002), with b-files by Schwob for $n = 0, \dots, 39$ and $n = 1, \dots, 37$; offsets 0 and 1, as the source says. None of the four entries has an asymptotic formula or a conjecture. *Checks*. The b-file terms for $n \leq 14$ equal the write’s Kostka computation, all terms of A178718 equal those of A104779, and the values for $n \leq 39$ and $n \leq 37$ were used in the comparisons of Section 1.2. No OEIS edit is part of this work.

11 Further questions

The support method leaves several concrete directions open.

Effective asymptotic remainders. The coefficient tails are explicit, but the global support and analytic stabilization estimates have not been optimized into useful numerical bounds on $|\log a_n - F_K(n)|$. Obtaining such bounds, with a practical starting index, would turn the two-ceiling theorem into a certified finite-input inverse algorithm. A leading numerical fit or a high-precision constant table is insufficient for this purpose.

Beyond fixed algebraic order. Appendix A supplies an exact two-sector decomposition, with a separately controlled recessive series smaller by $e^{-2\sqrt{n}}$. It does not describe optimal truncation or all exponentially small effects within either sector. Singularities governing coefficient stabilization may create additional smaller scales. A complete transseries treatment would require further sector definitions and sharper error estimates. The available remainder bound from a fixed finite dominant truncation is larger than the recessive scale, so that bound alone cannot isolate the smaller contribution numerically.

Higher profile corrections. The first signed density correction is the simple statistic $T_3 - H_3$. Further fixed supports should produce explicit polynomial statistics of the independent geometric margin variables. A systematic L^1 expansion would refine probabilities of profile events, while moment convergence for the dimension deficit requires its own uniform-integrability argument.

The cost of high order. The truncated multivariate product provides a finite coefficient algorithm, but partition growth makes a direct implementation expensive. Recurrences exploiting the logarithm of the product, weighted partition structure, and the root-count factorization may substantially reduce the cost. Sharp growth estimates for H_s and $J_s^\pm$ would also clarify the useful range of truncation.

Other matrix conventions. Removing weak margin order, prescribing the number of rows, or identifying equal-margin permutations changes the count and can change its scale. Those models require their own source identities and normalization checks. In particular, the binary-subset probability in (9.11) belongs only to the common variable-dimension sorted-margin universe specified here.

The central mechanism is robust: isolate the unit-cycle pole, prove convergence of each fixed marked coefficient, and control the entire moving-support tail separately. Its strength here is that it yields a rigorous expansion for an aggregate with unbounded dimension while keeping coefficient computation and inverse uncertainty explicit.

[write] **Added 7 October 2026, under Vladimir’s standing rule of 4 October 2026.** These five directions are the source’s own and stay open. The write adds from the non-claims a comparison with the journal version of Schwob’s paper and a proof of the imported identities independent of it (the write’s finite check reaches $n = 11$). No claim of the source was found false, and the OEIS entries contain no conjecture.

A An exact dominant and alternating sector decomposition

The fixed-order expansion can be refined without claiming a complete transseries or resurgence theory. The normal-moment representation of involution numbers admits an exact split over the two half axes. Each half has its own fixed-order asymptotic expansion, and the global support method transfers those separate expansions to the Kostka sum.

A.1 Exact sector definitions

For $\sigma \in \{+1, -1\}$, define

$$\mathcal{M}_m^\sigma = \frac{e^{-1/2}}{\sqrt{2\pi}} \int_0^\infty x^m e^{-x^2/2 + \sigma x} dx, \quad m \geq 0. \quad (\text{A.1})$$

If X is normal with mean 1 and variance 1, its moment generating function is $e^{z+z^2/2}$. Thus

$$I_m = \mathbb{E}X^m = \mathcal{M}_m^+ + (-1)^m \mathcal{M}_m^-. \quad (\text{A.2})$$

Write

$$w_{n,s} = \sum_{\substack{\rho \vdash s \\ \min \rho \geq 2}} \alpha_{1^{n-s}\rho} r(\rho), \quad \mathcal{A}_n^\sigma = \sum_{s=0}^n \sigma^s w_{n,s} \mathcal{M}_{n-s}^\sigma.$$

The empty type is included. Each $w_{n,s}$ is nonnegative, but $\mathcal{A}_n^-$ is defined by a signed sum. Inserting (A.2) in (2.2) gives the exact identity

$$a_n = \mathcal{A}_n^+ + (-1)^n \mathcal{A}_n^- \quad (n \geq 0). \quad (\text{A.3})$$

The two sectors are canonical relative to this specified half-axis split; no positivity of every recessive-sector summand is asserted.

Let

$$B_\sigma(n) = \frac{1}{\sqrt{2}} \left(\frac{n}{e}\right)^{n/2} e^{\sigma\sqrt{n}-1/4}, \quad \mathcal{K}(t) = \sum_{j \geq 0} k_j t^j$$

be the formal full Kostka series normalized by $CB_+(n)$. Its coefficients through k_6 are those in (6.2); in particular $k_0 = 1$, $k_1 = 7/24$, and $k_2 = -119/1152$.

Theorem A.1. *For every fixed $S \geq 0$ and either sign,*

$$\mathcal{A}_n^\sigma = C \sum_{s=0}^S \sigma^s H_s \mathcal{M}_{n-s}^\sigma + O_S(\mathcal{M}_n^\sigma n^{-(S+1)/2}). \quad (\text{A.4})$$

For every fixed $K \geq 0$, the separate sector expansions are

$$\mathcal{A}_n^\sigma = C B_\sigma(n) \left(\sum_{j=0}^K k_j \sigma^j n^{-j/2} + O_K(n^{-(K+1)/2}) \right). \quad (\text{A.5})$$

In particular, $\mathcal{A}_n^-$ is eventually positive, and

$$\frac{\mathcal{A}_n^-}{\mathcal{A}_n^+} = e^{-2\sqrt{n}} \left(1 - \frac{7}{12\sqrt{n}} + \frac{49}{288n} + O(n^{-3/2}) \right). \quad (\text{A.6})$$

The exact half-axis representation is a specialization of classical parabolic-cylinder integrals. In the notation of DLMF [9, (12.5.1)],

$$\mathcal{M}_n^\sigma = \frac{n! e^{-1/4}}{\sqrt{2\pi}} U(n + \tfrac{1}{2}, -\sigma). \quad (\text{A.7})$$

Olver's separate large-order expansions cover these two branches [10, §11]. The purpose of the elementary proof below is the transfer through the Kostka support weights, with precise normalization and remainder scope; it is not a novelty claim for the classical involution sectors.

A.2 Uniform half axis moment ratios

Let $R_k^\sigma = \mathcal{M}_k^\sigma / \mathcal{M}_{k-1}^\sigma$ for $k \geq 1$. Moment log-convexity from Cauchy–Schwarz and integration by parts give

$$R_{k+1}^\sigma \geq R_k^\sigma, \quad R_{k+1}^\sigma = \sigma + \frac{k}{R_k^\sigma} \quad (k \geq 1). \quad (\text{A.8})$$

For the minus sign, these imply $(R_k^-)^2 + R_k^- \leq k$, hence $R_k^- \leq \sqrt{k}$. For $k \geq 2$ the recurrence then yields

$$R_k^- \geq \sqrt{k-1} - 1 \geq \tfrac{1}{2}\sqrt{k} \quad (k \geq 9).$$

The finitely many earlier ratios are strictly positive. For the plus sign, $(R_k^+)^2 - R_k^+ \leq k$ gives $R_k^+ \leq 1 + \sqrt{k}$, and hence

$$R_k^+ \geq 1 + \frac{k-1}{1+\sqrt{k-1}} \geq \sqrt{k-1} \quad (k \geq 2).$$

Together with the positive first ratios, there is an absolute $c_* > 0$ such that $R_k^\sigma \geq c_* \sqrt{k}$ for both signs and all $k \geq 1$. Therefore

$$\frac{\mathcal{M}_{n-s}^\sigma}{\mathcal{M}_n^\sigma} \leq c_*^{-s} (n)_s^{-1/2} \leq \left(\frac{\sqrt{e}}{c_* \sqrt{n}} \right)^s. \quad (\text{A.9})$$

Using (3.3) and (3.5), even the absolute signed-sector support contribution is bounded by

$$w_{n,s} \frac{\mathcal{M}_{n-s}^\sigma}{\mathcal{M}_n^\sigma} \leq C_0 \left(4\sqrt{2e} c_*^{-1} n^{-1/4} \right)^s.$$

The tail from T is $O_T(n^{-T/4})$. Fixed-support stabilization still gives $w_{n,s} = CH_s + O_{s,R}(R^{-n})$. Splitting omitted supports at $T = 2S + 2$ proves (A.4), exactly as in Section 3, without relying on cancellation.

A.3 A real saddle and coefficient parity

Set $N = \sqrt{n}$, $t = 1/N$, and substitute $x = N + y$ in (A.1). Exact simplification gives

$$\begin{aligned} \frac{\mathcal{M}_n^\sigma}{B_\sigma(n)} &= \frac{1}{\sqrt{\pi}} \int_{-N}^{\infty} e^{-(y-\sigma/2)^2} e^{\Psi_N(y)} dy, \\ \Psi_N(y) &= N^2 \log(1 + y/N) - Ny + y^2/2. \end{aligned} \quad (\text{A.10})$$

Locally, for $|ty| < 1$,

$$\Psi_N(y) = \sum_{m \geq 1} t^m v_m(y), \quad v_m(y) = \frac{(-1)^{m+1} y^{m+2}}{m+2}.$$

Let $V_0(y) = 1$ and $V_j(y) = j^{-1} \sum_{m=1}^j m v_m(y) V_{j-m}(y)$. Then the half-axis coefficients are the rational Gaussian moments

$$\alpha_j^\sigma = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-(y-\sigma/2)^2} V_j(y) dy. \quad (\text{A.11})$$

Every v_m has parity m , so $V_j(-y) = (-1)^j V_j(y)$. Reflection gives

$$\alpha_j^- = (-1)^j \alpha_j^+. \quad (\text{A.12})$$

The plus Gaussian moments are computed by replacing y^d with

$$\sum_{r=0}^{\lfloor d/2 \rfloor} \binom{d}{2r} (1/2)^{d-2r} \frac{(2r-1)!!}{2^r},$$

so this is again a finite rational algorithm at each fixed order.

For completeness, the local expansion has a controlled remainder. Put $f_N(y) = N^2 \log(1 + y/N) - Ny - y^2/2$, for $y > -N$. Then $f_N(0) = f'_N(0) = 0$ and

$$f''_N(y) = -\frac{N^2}{(N+y)^2} - 1 \leq -1, \quad f_N(y) \leq -y^2/2.$$

Consequently the exact normalized integrand outside $|y| \leq N^\varepsilon$ is bounded by a Gaussian tail after absorbing the linear term σy , for fixed $0 < \varepsilon < 1/3$. On the central interval, Taylor's theorem through order K bounds the remainder by t^{K+1} times a fixed polynomial in $|y|$ times $e^{\delta y^2}$, for arbitrarily small fixed $\delta > 0$ once N is large. Here $|ty| = o(1)$, fixed derivatives are polynomially bounded, and the perturbation is $O(t|y|^3) \leq \delta y^2$. Gaussian integration and extension of the polynomial integrals to $\mathbb{R}$ give

$$\mathcal{M}_n^\sigma = B_\sigma(n) \left(\sum_{j=0}^K \alpha_j^\sigma n^{-j/2} + O_K(n^{-(K+1)/2}) \right). \quad (\text{A.13})$$

The leading terms show $\mathcal{M}_n^-/\mathcal{M}_n^+ \sim e^{-2\sqrt{n}}$. By (A.2) and uniqueness of the dominant Poincaré expansion, α_j^+ are exactly the coefficients α_j in (5.7).

A.4 Composition and the separate remainders

Let $\mathcal{I}(t) = \sum_{j \geq 0} \alpha_j t^j$, and define

$$E_s(t) = \frac{t^{-2} - s}{2} \log(1 - st^2) + \frac{s}{2} + \frac{\sqrt{1 - st^2} - 1}{t}.$$

The apparent singularities are removable. For the plus sector, the coefficient rule is

$$\mathcal{K}(t) = \sum_{s \geq 0} H_s t^s e^{E_s(t)} \mathcal{I}\left(\frac{t}{\sqrt{1 - st^2}}\right). \quad (\text{A.14})$$

Only $s \leq K$ enters a coefficient of degree K , so this is finite at every requested order. In the minus sector, the signed support weight $(-1)^s$, the reversed sign in the square-root exponential, and (A.12) combine to replace t by $-t$ in (A.14). Using (A.4) at $S = K$ and the finitely many shifted versions of (A.13) proves (A.5), with its separate relative error for each sector. Dividing the two expansions gives (A.6), since

$$\frac{\mathcal{K}(-t)}{\mathcal{K}(t)} = 1 - 2k_1 t + 2k_1^2 t^2 + O(t^3).$$

This proves an exact alternating contribution $(-1)^n \mathcal{A}_n^-$ with its own all-fixed-order expansion. A fixed finite dominant truncation provides only a remainder bound of order $B_+(n)n^{-(K+1)/2}$, which is larger than $B_-(n)$. That available bound does not isolate the recessive term by subtraction from a_n ; it is not a lower bound on every actual truncation error. The theorem neither classifies all smaller effects inside the exact sectors nor establishes optimal truncation, Stokes multipliers, or global resurgence.

[write] The two sectors at small n , 7 October 2026. For $15 \leq n \leq 39$ the write compared a_n (b-file) with $CB_+(n)\mathcal{K}(t)$ truncated after t^6 , with the constants above. Without the recessive sector the remainder, times $n^{7/2}$, alternates with the parity of n ($-14.3, -17.3$ at $n = 38, 39$); adding $(-1)^n CB_-(n)\mathcal{K}(-t)$, as (A.3) and (A.5) prescribe, removes the alternation ($-15.7, -15.9$). The alternating sector, of relative size $e^{-2\sqrt{n}}$, is thus already visible in the exact values; this is an observation, not a certificate. [Independent check, 7 October 2026. The numbers were reproduced, and a precision is needed. With the recessive sector added, the scaled remainder increases monotonically for $15 \leq n \leq 34$. For $34 \leq n \leq 39$ a residual alternation of about ± 0.1 remains ($-15.79, -15.87, -15.61, -15.79, -15.75, -15.98$), down from about ± 1.7 .

The residual comes from a third exponentially small effect, which both expansions omit because it lies beyond all their orders. The support weight $\alpha_{1^n} = [x^n] \prod_{j \geq 1} (1 - x^j/j!)^{-1}$ tends to C with error $O(2^{-n/2})$, from the poles $x = \pm\sqrt{2}$ of the factor $j = 2$. The pole at $-\sqrt{2}$ gives a_n an alternating relative term of about $0.0644 (-1)^n 2^{-n/2}$, comparable with $e^{-2\sqrt{n}}$ near $n = 33$ and larger below.

In units of $n^{-7/2}$, the exact correction $(\alpha_{1^n} - C)I_n/(CB_+(n))$ is -92.8 at $n = 15$ and -0.63 at $n = 39$. Subtracting it as well leaves a remainder that rises from -51.9 at $n = 15$ to about -13.0 at $n = 28$ – 30 and falls to -15.3 at $n = 39$. Its only parity wiggle is one of about 0.1 at the turning point, $n = 28$ – 30 .]

References

- [1] The OEIS Foundation Inc., *A104779: Sum of all the entries in the Kostka matrix of order n* , <https://oeis.org/A104779>, accessed 5 October 2026.
- [2] The OEIS Foundation Inc., *A178718*, official duplicate record of A104779, <https://github.com/oeis/oeisdata/blob/main/seq/A178/A178718.seq>, accessed 5 October 2026.

- [3] The OEIS Foundation Inc., *A321652: Nonnegative integer matrices with weakly decreasing row and column sums*, <https://oeis.org/A321652>, accessed 5 October 2026.
- [4] The OEIS Foundation Inc., *A068313: Binary matrices with nonzero weakly decreasing row and column sums*, <https://oeis.org/A068313>, accessed 5 October 2026.
- [5] L. Schwob, *On the enumeration of double cosets and self-inverse double cosets*, arXiv:2506.04007v1, 4 June 2025, <https://arxiv.org/abs/2506.04007>; inspected full text: <https://arxiv.org/html/2506.04007v1>. Published in *Advances in Applied Mathematics* **173** (2026), 102982, <https://doi.org/10.1016/j.aam.2025.102982>. The journal full text was inaccessible; claims of source comparison concern the inspected arXiv version.
- [6] L. Moser and M. Wyman, *On solutions of $x^d = 1$ in symmetric groups*, *Canadian Journal of Mathematics* **7** (1955), 159–168, <https://doi.org/10.4153/CJM-1955-021-8>.
- [7] L. Moser and M. Wyman, *Asymptotic expansions*, *Canadian Journal of Mathematics* **8** (1956), 225–233, <https://doi.org/10.4153/CJM-1956-026-x>.
- [8] F. Bornemann, *Asymptotic expansions relating to the lengths of longest monotone subsequences of involutions*, arXiv:2306.03798v5, Appendix C, <https://arxiv.org/html/2306.03798v5#A3>; published article: <https://doi.org/10.1080/10586458.2024.2397334>. The involution-count expansion used here does not require the separate distributional hypotheses elsewhere in that paper.
- [9] NIST Digital Library of Mathematical Functions, *Parabolic cylinder functions*, §12.5, equation (12.5.1), and §12.10(v), <https://dlmf.nist.gov/12.5.E1>, <https://dlmf.nist.gov/12.10.v>, accessed 5 October 2026.
- [10] F. W. J. Olver, *Uniform asymptotic expansions for Weber parabolic cylinder functions of large orders*, *Journal of Research of the National Bureau of Standards B* **63B** (1959), 131–169, <https://doi.org/10.6028/jres.063B.014>, especially §11.
- [11] ProveIt, *Transseries and inversion*, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; Part 0 (`p0:def:model`, `p0:thm:lambert-core`, `p0:def:three-inverses`, `p0:thm:staircase`) and the chapter on the Lambert template (`plt:def:lw-monomial-log-datum`, `plt:thm:lw-template`). [Added in the write.]